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Relationship Between Degree of Stenosis and Time-To-Peak Delay in High Grade Asymptomatic Carotid Artery Disease

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Short title: Asymptomatic Carotid Stenosis and TTP Delay

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ABSTRACT

Background: Cerebral hemodynamic impairment in patients with asymptomatic high-grade (>70%) internal carotid artery (ICA) stenosis is associated with risk of stroke and cognitive decline, but correlations between degree of stenosis and hemodynamic impairment are derived from small case series. Using baseline data from 242 participants in the Carotid Revascularization and Medical Management for Asymptomatic Carotid Stenosis–Hemodynamics (CREST-H) study, we hypothesized that degree of stenosis in the ICA would correlate with time-to-peak (TTP) delay in ipsilateral cerebral blood flow using MR/CT perfusion scans, adjusting for demographic and cardiovascular risk variables.

Methods: From all participants, ICAs were insonated using a 7-10MHz probe for peak systolic velocity (PSV) and end diastolic velocity (EDV). Dynamic contrast perfusion imaging was standardized across 61 CREST-H sites, using standard sequential T2*-weighted perfusion imaging. CT perfusion used standard clinical protocols. TTP delay was calculated in the ipsilateral versus contralateral hemispheres. In cross-sectional analysis, linear regression used TTP delay as the outcome variable and PSV, EDV, and PSV ICA/CCA ratio (ICR) on the index and non-index side as primary predictor variables in 6 separate models, adjusting for covariates, followed by automated backward elimination model reduction.

Results: Among 392 CREST-H participants, the 242 with complete data for all variables were included in the regression analysis (Age 70±7.6, 62%M). EDV on the index side correlated with TTP delay (β=0.003, p=0.005). PSV had a similar correlation but did not reach significance (β=0.001, p=0.099), nor did ICR (β=0.003, p=0.126). Non-index side PSV, EDV, and ICR showed no correlation with TTP delay (p=0.268, p=0.495, p=0.380 respectively). Circle of Willis completeness did not correlate with TTP.

Conclusions: In this large cohort of patients with high-grade asymptomatic carotid stenosis, higher end-diastolic flow velocities correlated with greater TTP delays, supporting a definition of

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"hemodynamically significant stenosis." Our findings have implications for management of asymptomatic carotid stenosis which will be tested in the CREST-H study.

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Non-standard Abbreviations and Acronyms.

CoW = Circle of Willis

EDV = End diastolic velocity

ICA = Internal carotid artery

ICR = Internal Carotid Artery/Common carotid artery ratio

IMM = intensive medical management

PSV = Peak systolic velocity

TTP = Time-to-peak

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INTRODUCTION.

Cerebral hemodynamic impairment in high-grade internal carotid artery (ICA) stenosis is associated with increased risk of stroke[1,2] and cognitive impairment[3–6], even in the absence of cerebral infarction or stroke symptoms. Although hemodynamic impairment may have implications for management of asymptomatic carotid stenosis, the term “hemodynamically significant stenosis” is widely used but poorly defined. High-grade carotid stenosis can reduce cerebral blood flow in the hemisphere ipsilateral to the stenosis by 8 to 24%[7,8] and delay the arrival time of contrast material to the hemisphere supplied by the stenotic carotid artery in up to 50% of patients.[9,10] Yet it remains uncertain how well the degree of stenosis in the ICA correlates with degree of hemodynamic impairment in the ipsilateral hemisphere. Current descriptions of the relationship between degree of carotid stenosis and TTP delay are derived from small case series[11–17] which has not allowed adjustment for additional variables that could influence cerebral blood flow. For this report, we used adjudicated baseline data from the Carotid Revascularization and Medical Management for Asymptomatic Carotid Stenosis – Hemodynamics (CREST-H) study[18]. We hypothesized that degree of stenosis in the ipsilateral ICA would correlate with time-to-peak (TTP) delay using MRI or CT perfusion scans, while adjusting for completeness of the circle of Willis and 11 demographic and vascular risk variables which have reported associations with brain perfusion in the literature and were included in the analysis.

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METHODS.

Participants.

CREST-H is an NINDS-funded clinical investigation that is an ancillary study to the NINDS-funded Carotid Revascularization and Medical Management for Asymptomatic Carotid Stenosis

Study (CREST-2) randomized controlled trial.[18] CREST-2 is a pair of outcome-blinded, phase 3 randomized controlled trials to assess treatment differences in stroke and death rates in patients assigned to: 1) Carotid artery stenting plus intensive medical management (CAS+IMM) versus IMM alone, and 2) Carotid endarterectomy plus IMM (CEA+IMM) versus IMM alone.[19] Participants randomized into the CREST-2 parent study and enrolled in CREST-H between 01/30/2018 and 07/03/2024 were eligible for inclusion in this study.

Per the CREST-2 protocol, participants had ≥70% internal carotid artery (ICA) stenosis and no symptomatic stroke or transient ischemic attack in the target carotid territory within the prior 180 days. Stenosis was defined by carotid duplex ultrasound using a 7-10MHz probe, showing a Doppler peak systolic velocity (PSV) ≥230cm/sec plus one of the following additional findings: end diastolic velocity (EDV) ≥100cm/sec, PSV ICA/CCA ratio (ICR) ≥4.0, or CTA/MRA demonstrating ≥70% stenosis, or at least 70% stenosis on digital subtraction angiography (DSA) alone.[20] All ultrasound labs were credentialed for the study, and Doppler images were adjudicated centrally at the CREST-2 Imaging Core at the University of Maryland. Additional eligibility criteria for CREST-H included age 18-86 years, fluency in English, no contralateral stenosis ≥70% or occlusion, no pre-existing diagnosis of dementia, no history of severe head trauma or current major depression, and education ≥8 years.[18] Written informed consent based on IRB approval from a central Institutional Review Board (IRB) at University of Cincinnati or local site IRB was obtained from all participants, first for CREST-2 and then separately for CREST-H. The current study was approved by the Columbia University Irving Medical Center IRB, according to the ethical standards of Columbia University under the Common Rule as revised January 21, 2019. Deidentified data are available from the corresponding author on reasonable request. The study is reported according to STROBE criteria.

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<u>Imaging acquisition.</u>

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MRI Dynamic Susceptibility Contrast (DSC) perfusion scans (N=236), obtained on 1.5T or 3T scanners, or CT perfusion scans (N=6) obtained on clinical scanners, were acquired at each participating site according to parameters established in the CREST-H study protocol and were processed centrally at the imaging core lab at UCLA Geffen School of Medicine. The MRI protocol included DSC, T2-weighted fluid attenuated inversion recovery (FLAIR) images to assess infarcts and white matter hyperintensities (WMH), time of flight angiography for circle of Willis (CoW) assessment, and high resolution T1-weighted images, whereas the CT protocol included non-contrast imaging and a CT perfusion (CTP) sequence. MRI DSC sequences were tailored for each participating site's standard MRI vendor system and field strength. For example, GE 3T DSC parameters included: flip angle=60 deg, phase/loc=128, slice thickness 5.0mm, gap=0, slices=max for TR, TR=1500ms. TE=30ms, FOV=22cm, frequency=96, phase=128, mode=2D. Protocol-specified CTP imaging was alternatively acquired at each site, similar or identical to clinical protocols for the site's acute stroke CTP imaging. DSC and CTP source images were post-processed with a semi-automated system that computed quantitative perfusion maps using deconvolution of tissue and arterial signals (Olea, Cambridge, MA), yielding standardized data regardless of the acquisition system at each site. The relative TTP delay was represented as a difference TTP (dTTP) calculated by subtraction of voxel values in the hemisphere ipsilateral minus those contralateral to the stenosis, using OleaSphere 3.0 software, with the TTP delay measured in the middle, anterior, and posterior cerebral artery territories. TTP delay thus reflects any extracranial or intracranial proximal stenosis, as well as variations in circle of Willis collateral anatomy. We used the whole hemisphere to assess hemodynamic asymmetry since ICA stenosis may affect blood flow in the anterior brain regions directly and posterior regions indirectly.[8,21] We chose TTP delay as the hemodynamic measure for this study because of its high sensitivity to interhemispheric asymmetry, reproducibility, and availability across different scanners and field strengths.[22] TTP delay in carotid artery disease has been shown to correlate with other perfusion-related measures such as increased mean

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transit time (MTT) and time to maximum (Tmax) and does not require an arterial input function.[9,23] The computer algorithm binned dTTPs into thresholds of 0.25-second increments up to ≥2 seconds, except for 0-0.5 seconds, the only interval requiring a greater delay range for reliable measurements distinguishing asymmetry from no asymmetry by the automated algorithm. To be classified as asymmetrical, a minimum volume of 10cc was required at each level of relative TTP delay. As a result, there were 8 gradations of asymmetry, ranging from no asymmetry (TTP delay=0.0-0.49 seconds) to a TTP delay of ≥2 seconds. Assessors were blinded to degree of stenosis. Circle of Willis segments -- proximal index side and non-index side anterior cerebral arteries (A1 segments), anterior communicating artery (ACom), index and non-index side proximal posterior cerebral arteries (P1 segments), and index side and non-index side posterior communicating arteries (PCom) – were assessed by 2 experienced neuroradiologists (JH, PR) for presence or absence of each segment. Anterior circulation completeness was defined as presence of both A1s and Acom; posterior circulation completeness was defined as presence of P1 and Pcom on the index side. Collateral segments were classified as incomplete if any of the segments were absent.

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Statistical Analysis.

From the 392 total subjects, we excluded 150 subjects with missing data on one or more covariates. There was no significant difference in covariates between the participants included in the analysis and those excluded (see **Supplemental Material Table S1**). We used Pearson's correlation to examine the association between TTP delay and degree of stenosis in the Index (target artery) and non-index (contralateral) ICA and ran adjusted linear regression models with significance level set at α=0.05 for our primary analysis. We tested six separate models to assess the relationship of the 3 measures of luminal stenosis by carotid ultrasound (PSV, EDV, ICR) for the index side and non-index side carotid artery with TTP delay. TTP delay was

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designated as the outcome variable, and the carotid Doppler flow velocity measures as the primary predictors, adjusting for age, sex, education level (less than high school, high school grad, some college, college grad (reference group)), history of hypertension, history of diabetes, low density lipoprotein (LDL) level, history of prior stroke, smoking history, coronary artery disease, left vs right hemisphere, exercise level (regular vigorous exercise, regular moderate exercise (reference group), irregular moderate/vigorous exercise, no moderate/vigorous exercise), and presence of a complete versus incomplete anterior CoW, posterior CoW, or both anterior and posterior CoW. Our covariates were chosen from data available in the CREST-2 data set, based on demographic and clinical factors that could contribute to altered cerebral hemodynamics, independently of carotid stenosis. All covariates were kept in the fully adjusted models, followed by an automated stepwise backward elimination model-reduction algorithm that set entry criteria at p=0.15, and retention level at p=0.05.

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Results.

A total of 392 participants were enrolled in CREST-H across 61 participating sites; 63% were men and 33% had less than college education. All had unilateral carotid stenosis >70%. The 242 participants with complete data for all variables were included in the regression analysis. Patient characteristics are shown in **Table 1**. In univariate correlations, index side EDV but not PSV or ICR correlated with TTP delay (r= 0.16, p=0.011; r=0.10, P=0.125; r=0.06, p=0.369 respectively). Univariate correlations between Doppler stenosis measures and TTP delay are shown in **Table 2** and represented in **Figure 1.**

In the fully adjusted model using EDV on the index side and all covariates, EDV (in cm/sec) correlated with TTP delay (β=0.003, p=0.005) (**Table 3**). In the index-side PSV model, PSV did not reach statistical significance in the regression model (β=0.001, p=0.099) (**Table 4**), nor did

ICR (β=0.003, p=0.126) (**Table 5**). Non-index side PSV, EDV, and ICR did not correlate with TTP delay (p=0.268, p=0.495, p=0.380 respectively). **Supplemental tables S2-S4 in supplementary materials** show the full regression results for EDV, PSV, and ICR on the non-index side.

In our cohort, 73% had an intact anterior CoW, 49% had an intact posterior CoW, 18% had a complete CoW that included all 7 anterior and posterior segments, and 36% had a complete CoW for anterior segments and posterior segments on the index side. None of the CoW variables correlated with TTP (p=0.19, p=0.93, p=0.80, p=0.43 respectively for the EDV index side model). Compared with moderate exercise, self-reported vigorous exercise correlated with shorter TTP delay (p=0.007 for the EDV index side model). Compared with college graduates, those with HS education had longer TTP delays (p=0.011 for the EDV index side model). When the automated backward elimination algorithm was applied to each model, only the EDV model retained the luminal stenosis variable. EDV correlated with TTP delay (β=0.003, p=0.005), as did vigorous exercise (β= -0.544, (p=0.006) and high school education (β=0.334, p=0.011). (**Table 6**). (The backward elimination output by step is provided in **supplemental materials**.)

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Discussion.

In this large, well-characterized cohort of patients with unilateral, high-grade, asymptomatic carotid stenosis, we found that degree of stenosis in the ICA, measured by end diastolic flow velocity on Duplex ultrasonography, correlated with degree of TTP delay in the ipsilateral hemisphere. Our findings are consistent with existing descriptions in the literature,[11,24] but the large size of our study allowed us to adjust for 12 demographic, anatomical, and cardiovascular risk variables. This is the first study to use a fully adjusted model to describe the relationship

between a measure of carotid luminal stenosis and a direct measure of cerebral hemodynamic impairment.

Of the 3 measures of carotid stenosis, we found that only EDV correlated with TTP delay. Although PSV is the most commonly used measure to define degree of carotid stenosis, it did not correlate with our perfusion measure of TTP delay. The reason for this may have been the way it was used in the study design. As the primary eligibility criterion for inclusion in the CREST-2 clinical trial, PSV range was limited to ≥230cm/sec, creating a "floor effect" and thereby reducing the chance of identifying an association with other variables. PSV had a median of 347 cm/sec with an interquartile range (280-421cm/sec), with only 12 participants having baseline PSV <230cm/sec. EDV, by contrast, used in the study to confirm the degree of carotid stenosis once the PSV≥230cm/sec threshold inclusion criterion was met, had a median of 110cm/sec and an interquartile range 76-140cm/sec, with the distribution nearly evenly divided above and below the EDV threshold of 100cm/sec.

Complete versus incomplete CoW did not correlate with TPP, neither in the anterior collateral circuit across the anterior communicating artery nor in the posterior circuit via the ipsilateral P1 segment and posterior communicating artery. Despite being a backup system for impaired flow to the brain from stenosis of one or more precerebral arteries, a complete circle of Willis of all 7 segments is seen in as few as 10% in some populations.[25] In the current cohort, a complete CoW was present in 18%. CoW may not have correlated with TTP in our models because it is a conduit for collateral flow in a minority of patients.

Other covariates were included in the model because of purported contributions to cerebral perfusion. For example, CBF decreases with age,[26] women on average maintain a higher CBF than men[27], hypertension is associated with lower CBF[28,29] and "rightward" shift of the autoregulatory curve toward higher blood pressures[30], and chronic stroke reduces overall hemispheral blood flow[31]. Two of the covariates – exercise and education – emerged as

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significant correlates with TTP in our regression models. Supporting evidence for the association of exercise with lower TTP delay includes a recent randomized clinical trial showing that older adults had increased CBF after a 1-year aerobic exercise training program.[32] Meta-analyses have also shown correlations between physical exercise and higher CBF.[33,34] Reasons for the correlation between lower education level and longer TTP delay is less clear. Higher education has been shown to correlate with higher CBF in certain brain regions in patients with Alzheimer's disease,[35,36] but its impact in patients with high-grade carotid stenosis patients has not been investigated.

This study has limitations. First, 150 CREST-H participants were excluded due to one or more missing data elements. The majority comprised missing or uninterpretable perfusion scans (N=80) or Doppler data (N=34) provided by our CREST-H participating sites. We chose to use only participants with complete data sets for this physiological investigation, rather than impute missing data. Although we were able to confirm no clinically significant difference in baseline characteristics between those excluded versus not excluded, the lower N reduced our statistical power. Second, our measurement of carotid stenosis relied solely on Doppler variables, of which only EDV correlated with TTP. Although PSV is generally used as a standard for defining stenosis, our PSV values were restricted at the lower end by CREST-2 entry criteria, reducing chances to identify linear correlations. Furthermore, although use of CTA, MRA or DSA measurements could help validate our findings, CTA and MRA results from participating sites included only whether the stenosis was >70%; therefore there was insufficient information to run a linear regression analysis with these data. In addition, the number of participants in the trial who used DSA for eligibility was too small to allow analysis of correlations between the vascular imaging results and TTP delay. Third, although CREST-H excluded contralateral stenosis >70%, contralateral stenosis of 50-69% could theoretically have affected the TTP delay results. We ran the same regression analysis, eliminating participants with contralateral stenosis 50-

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69%; it showed no difference in the results (**Table S5** in **supplementary material**). Fourth, Doppler measurement began at study entry and therefore we cannot comment on TTP association with duration of stenosis. Fifth, choice of covariates was limited to those available in the CREST-2 parent trial. Other variables that might affect cerebral hemodynamics such as PCO2 levels, caffeine use, or medications, were not included. It is possible that inclusion of such variables could have altered the correlation between the Doppler variables and TTP delay. Finally, CoW measurements were limited to anatomical visualization of segment lumens. We did not have directional flow information in the arteries to know whether CoW collaterals were rerouting flow, even if the segments appeared patent. Furthermore, the circle of Willis is not a static system. Segmental lumen diameters may change as flow demand changes. Planned longitudinal analysis from the CREST-H study may allow further examination of how the CoW changes over time, including before and after revascularization of the target ICA.

Even with the limitations, the demonstration of a correlation between degree of carotid stenosis, specifically measured by EDV, and TTP delay supports the fundamental principle of a "hemodynamically significant" stenosis: for carotid stenosis above 70%, the higher the degree of narrowing, the greater the hemodynamic impairment, independent of other contributory variables. Our findings lay the groundwork for the investigation of the impact of revascularization on blood flow and cognition as planned in the CREST-H study. The relationship between exercise, education and cerebral hemodynamics also warrants further investigation.

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Tables

Table 1. Patient characteristics of the 242 participants analyzed.

Variable	Average ± SD or Prevalence N (%)
Age (mean±SD)	70.1 ± 7.6
Sex (M)	150 (62%)
Educ 1 <High school	24 (10%)
Educ 2 High school grad	61 (25%)
Educ 3: some college	82 (34%)
Educ 4:college graduate	75 (31%)
Hypertension	201 (83%)
Diabetes	77 (32%)
LDL (mean±SD)	75.7 ± 29.2
History prior stroke	56 (23%)
Smoking (ever smoked)	177 (73%)
Coronary artery disease	94 (39%)
Index artery side	116 (48%)
Exercise 1: regular vigorous exercise	17 (7%)
Exercise 2: irregular moderate/vigorous exercise	77 (32%)
Exercise 3: no moderate/vigorous exer	61 (25%)
EDV index side (mean±SD)	113.2 ± 49.6

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PSV index side (mean±SD)	358.5 ± 111.4
Ratio index side (mean±SD)	6.2 ±5.8
EDV non-index side (mean±SD)	44.0 ± 29.1
PSV non-index side (mean±SD)	148.8 ± 87.7
Ratio non-index side (mean±SD)	2.0 ± 1.7
Posterior collateral	53%
Anterior collateral	74%
Both anterior and posterior collateral	19%
Both ant and post collateral -- index	38%

LDL=low density lipoprotein, educ=education, EDV=end diastolic velocity, PSV=peak systolic velocity, index side refers to the target carotid artery side (left or right) in CREST-2 eligibility

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Table 2. Correlations between TTP delay and Doppler measures of stenosis.

	ttp_delay	MaxIndEDV	MaxIndPSV	MaxIndRatio	MaxNIndEDV	MaxNIndPSV	MaxNIndRatio
ttp_delay	1.0000 242	0.1642 0.011 242	0.0989 0.125 242	0.0582 0.369 241	-0.0316 0.628 238	-0.0424 0.515 239	-0.0606 0.351 239
MaxIndEDV **MaxIndEDV**	0.1642 0.011 242	1.0000 242	0.7210 <.0001 242	0.4446 <.0001 241	0.0452 0.488 238	0.0002 0.998 239	-0.0837 0.197 239
MaxIndPSV **MaxIndPSV**	0.0989 0.125 242	0.7210 <.0001 242	1.0000 242	0.5186 <.0001 241	-0.0053 0.935 238	0.1441 0.026 239	-0.0782 0.228 239
MaxIndRatio **MaxIndRatio**	0.0582 0.369 241	0.4446 <.0001 241	0.5186 <.0001 241	1.0000 241	-0.0322 0.622 237	0.0104 0.873 238	-0.0370 0.570 238
MaxNIndEDV **MaxNIndEDV**	-0.0316 0.628 238	0.0452 0.48 238	-0.0053 0.935 238	-0.0322 0.622 237	1.0000 238	0.8334 <.0001 238	0.7324 <.0001 238
MaxNIndPSV **MaxNIndPSV**	-0.0424 0.515 239	0.0002 0.998 239	0.1441 0.026 239	0.0104 0.873 238	0.8334 <.0001 238	1.0000 239	0.8237 <.0001 239

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MaxNIndRatio	-0.0606	-0.0837	-0.0782	-0.0370	0.7324	0.8237	1.0000
MaxNIndRatio	0.351	0.197	0.228	0.570	<.0001	<.0001	
	239	239	239	238	238	239	239

Each cell contains the Pearson correlation coefficient, p-value, and number of observations used. MaxIndEDV=maximum end diastolic flow velocity on the index (target artery) side, MaxIndPSV=maximum peak systolic flow velocity on the index side, MaxIndRatio=maximum ICA/CCA ratio velocity on the index side, MaxNIndEDV=maximum end diastolic flow velocity on the non-index (contralateral) side, MaxNIndPSV= maximum peak systolic flow velocity on the non-index side, MaxNIndRatio= maximum ICA/CCA ratio velocity on the non-index side.

Table 3. The fully adjusted model of TTP delay regressed on covariates including index side EDV.

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	0.228	0.584	0.39	0.670
Age	0.007	0.006	0.97	0.335
Sex	0.058	0.102	0.56	0.574
Education 1: <high school	0.194	0.182	1.06	0.290
Education 2: high school grad	0.334	0.130	2.56	0.011*
Education 3: some college	-0.072	0.119	-0.61	0.545
Hypertension	-0.095	0.129	-0.74	0.460
Diabetes	-0.135	0.103	-1.31	0.193
LDL	-0.002	0.002	-0.88	0.379
History prior stroke	0.021	0.044	0.48	0.629
Smoking	-0.013	0.108	-0.12	0.905
Coronary artery disease	0.014	0.100	0.14	0.886
Index artery side	-0.134	0.100	-1.39	0.166
Exercise 1: regular vigorous	-0.544	0.200	-2.75	0.006**
Exercise 2: Irregular moderate/vigorous	-0.073	0.117	-0.63	0.531
Exercise 3: no moderate/vigorous	-0.029	0.125	-0.24	0.814
Maximum EDV Index side	0.003	0.001	2.85	0.005**
Posterior Collateral	-0.244	0.185	-1.31	0.191
Anterior Collateral	0.013	0.155	0.09	0.931
Both anterior and posterior collateral	-0.039	0.154	-0.25	0.802
Both anterior and posterior index side	0.179	0.228	0.78	0.434

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LDL=low density lipoprotein, Education values are referenced to college graduate, Exercise values are referenced to regular moderate exercise, EDV=end diastolic flow velocity on the index side, **=p<0.01, *=p<0.05

Table 4. The fully adjusted model of TTP delay regressed on covariates including index side PSV.

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	0.531	0.576	0.92	0.358
Age	0.003	0.007	0.44	0.660
Sex	0.060	0.104	0.58	0.565
Education 1: <high school	0.187	0.185	1.01	0.311
Education 2: high school grad	0.336	0.132	2.55	0.011
Education 3: some college	-0.071	0.121	-0.58	0.560
Hypertension	-0.134	0.129	-1.04	0.300
Diabetes	-0.161	0.104	-1.55	0.123
LDL	-0.002	0.002	-0.90	0.368
History prior stroke	0.021	0.045	0.47	0.637
Smoking	-0.023	0.109	-0.21	0.834
Coronary artery disease	0.003	0.100	0.03	0.976
Index artery side	-0.119	0.098	-1.22	0.224
Exercise 1: regular vigorous	-0.541	0.200	-2.70	0.007
Exercise 2: Irregular moderate/vigorous	-0.075	0.118	-0.63	0.526
Exercise 3: no moderate/vigorous	-0.028	0.126	-0.22	0.822
Maximum PSV Index side	0.001	0.0006	1.66	0.099
Posterior Collateral	-0.183	0.186	-0.98	0.326
Anterior Collateral	0.028	0.157	0.18	0.857
Both anterior and posterior collateral	-0.084	0.155	-0.54	0.590
Both anterior and posterior index side	0.155	0.231	0.67	0.502

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LDL=low density lipoprotein Education values are referenced to college graduate, Exercise values are referenced to regular moderate exercise, PSV=peak systolic flow velocity on the index side, **=p<0.01, *=p<0.05

Table 5. The fully adjusted model of TTP delay regressed on covariates including index side ICR.

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	0.670	0.571	1.17	0.242
Age	0.004	0.007	0.54	0.587
Sex	0.064	0.105	0.61	0.544
Education 1: <high school	0.211	0.185	1.14	0.256
Education 2: high school grad	0.356	0.132	2.70	0.008*
Education 3: some college	-0.055	0.121	-0.45	0.654
Hypertension	-0.121	0.131	-0.93	0.355
Diabetes	-0.163	0.105	-1.56	0.120
LDL	-0.002	0.002	-0.99	0.326
History prior stroke	0.023	0.045	0.50	0.615
Smoking	-0.029	0.110	-0.26	0.793
Coronary artery disease	0.007	0.101	0.07	0.946
Index artery side	-0.130	0.098	-1.33	0.186
Exercise 1: regular vigorous	-0.583	0.204	-2.85	0.005*
Exercise 2: Irregular moderate/vigorous	-0.080	0.119	-0.67	0.502
Exercise 3: no moderate/vigorous	-0.0298	0.127	-0.23	0.815
Maximum ICR Index side	0.013	0.009	1.54	0.126
Posterior Collateral	-0.162	0.186	-0.87	0.385
Anterior Collateral	0.022	0.158	0.14	0.891
Both anterior and posterior collateral	-0.071	0.157	-0.45	0.653
Both anterior and posterior index side	0.137	0.231	0.59	0.555

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Education values are referenced to college graduate, Exercise values are referenced to regular moderate exercise, ICR=internal carotid artery/common carotid artery ratio on the index side, LDL=low density lipoprotein, **=p<0.01, *=p<0.05

Table 6. Significant variables retained in the EDV model after automated backwards elimination algorithm was applied.

Parameter	**DF**	**Estimate**	**Standard Error**	**T Value**	**Pr > [t]**
Intercept	1	0.187	0.117	1.60	0.112
Education 2: high school grad	1	0.356	0.106	3.36	0.001
Exercise 1: regular vigorous	1	-0.4946	0.179	-2.76	0.006
Maximum EDV index side	1	0.003	0.001	2.81	0.005

Education values are referenced to college graduate, Exercise values are referenced to regular moderate exercise, ICR=internal carotid artery/common carotid artery ratio on the index side, LDL=low density lipoprotein

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Figure 1 shows a scatterplot with Doppler End Diastolic flow Velocity (EDV) in cm/sec on the x-axis and Time-to-peak (TTP) delay on the y-axis. EDV was shown to be significantly correlated with TTP delay (r=0.16, Beta=0.003, p=0.005).

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Supplemental Material

Tables S1-S5

Supplemental Methods – Model Reduction Steps

Checklist

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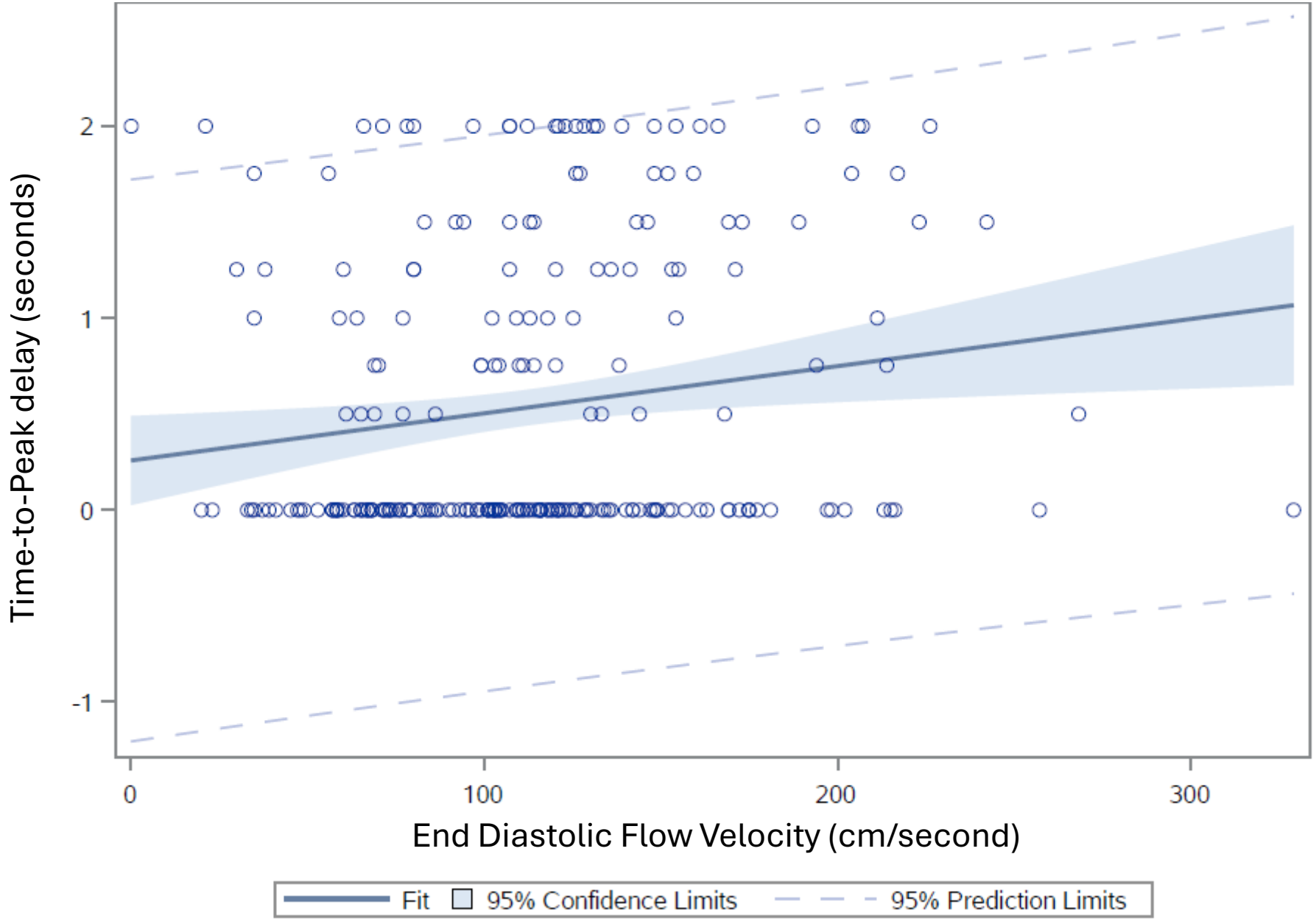
Figure 1. Scatterplot with Regression Line: End Diastolic Flow Velocity vs Time-to-Peak Delay



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